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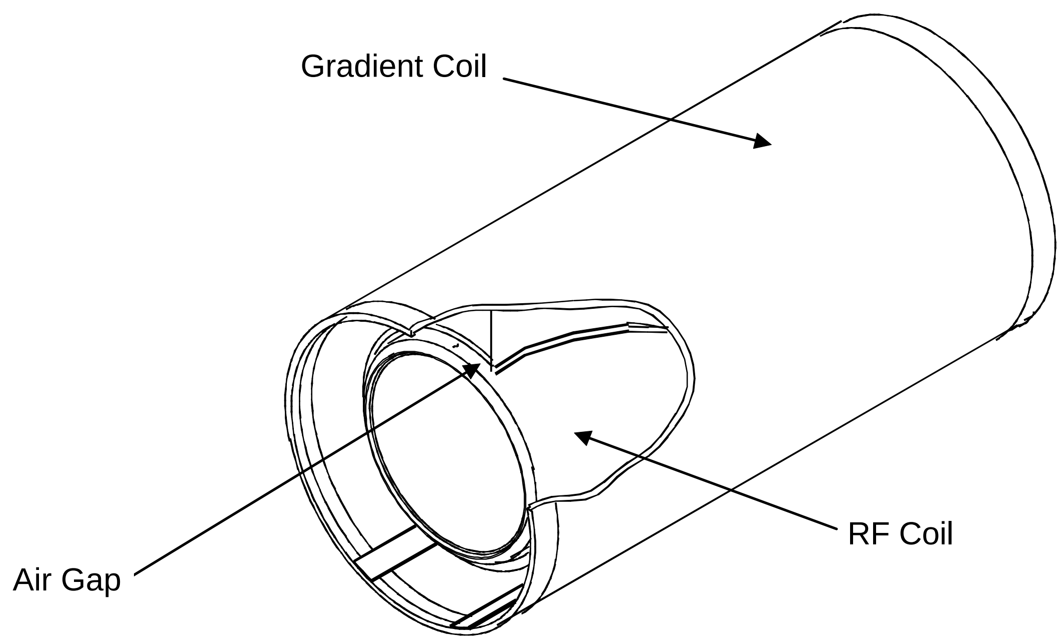
Description - This is a simple test to check the functionality of the Body Coil Air Blower system.

The Body Coil Air Blower system cools the patient bore wall surfaces by forcing air between the gradient coil and the RF coil. (See Illustration 1-1). The rear of the magnet needs to have an air-tight seal in order to maintain a sufficient pressure head to force air to flow, which transfers heat away from the bore wall with convection. If there is not an air tight seal in the rear of the magnet, there will not be enough pressure to force air through the space between the Gradient and RF coils.

Note

Air flow cooling is more critical in the Wide Open product than in previous products. Failure of the air flow cooling system may result in bore wall temperature sensor trips.



RF/GRADIENT COIL
ILLUSTRATION 1-1

1- PROCEDURE

The procedure consists of checking the most probable areas for air leakage by simply placing a hand in the area, and feeling for a **<u>LARGE</u>** leak. Some small leaks may exist. Use good judgment as to the size of the leak. A visual inspection of air seals will usually indicate if a seal has failed. If a seal has failed, check to make sure the seal is not out of position, or not under enough compression. The probable areas for leakage are as follows.

1. The flange joint between the RF Tube and the End Bell. (See Illustration 1-2)

2. The two holes in back which support the bridge. (See Illustration 1-2)

3. The I & Q RF and Bias lines. (See Illustration 1-2)

4. The air hose connection at the magnet. (See Illustration 1-2)

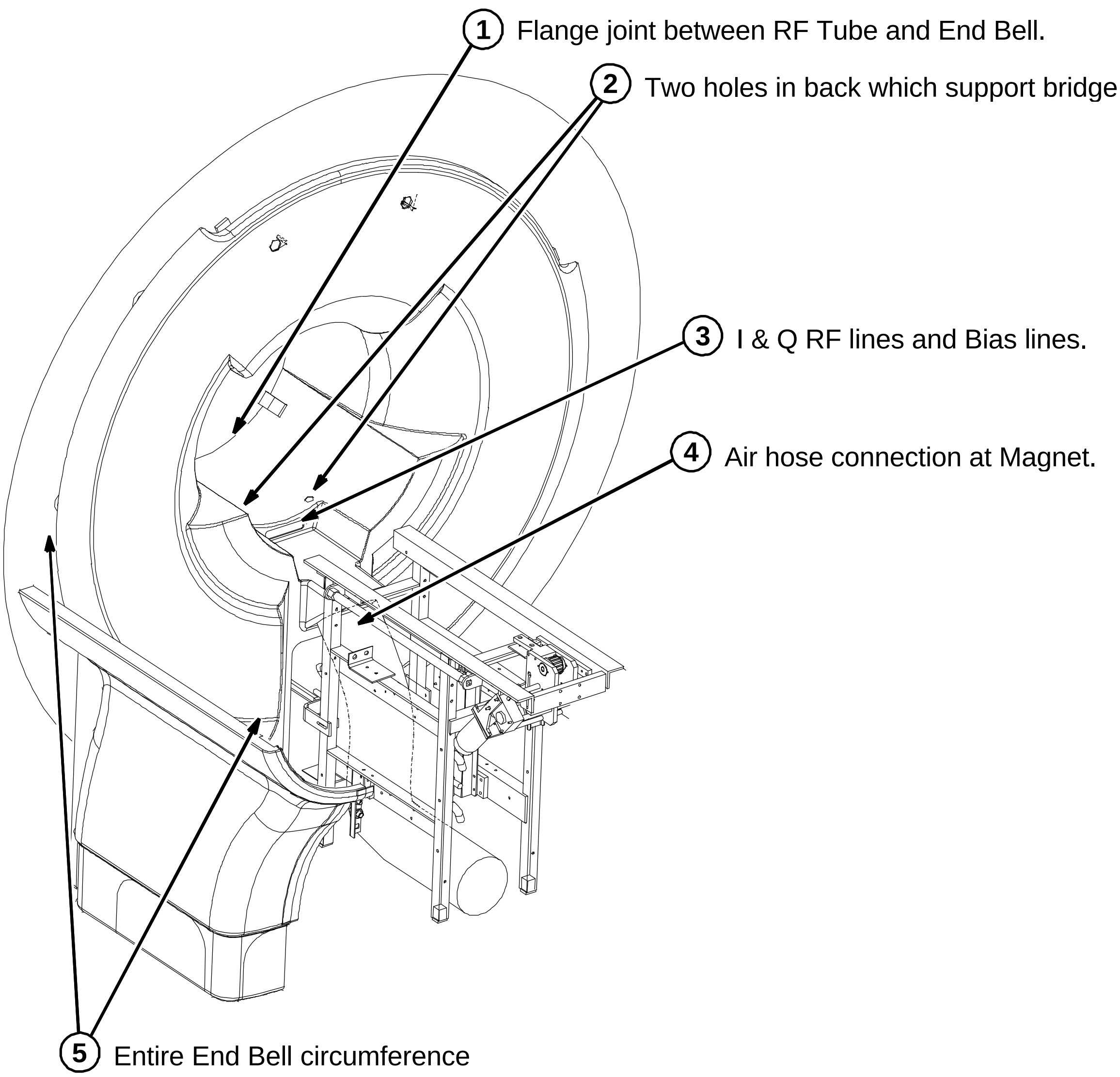
5. The entire End Bell circumference. This is an extremely important seal between the magnet and the End Bell. Inspect it carefully for compression of the seal. (See Illustration 1-2)

Note

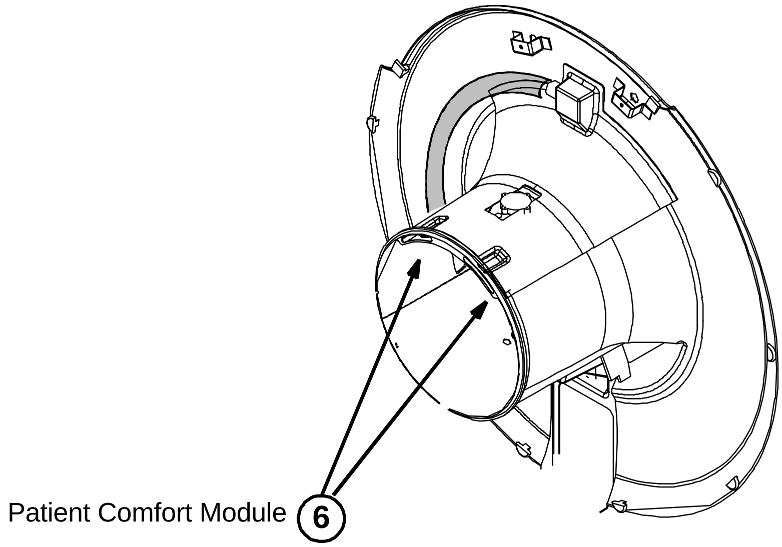
If the End Bell is secured tighter at the top than at the bottom, a leak may occur at the bottom of the End Bell.

6. The Patient comfort module should not have air blowing from it when the patient fan is off. (See Illustration 1-3)

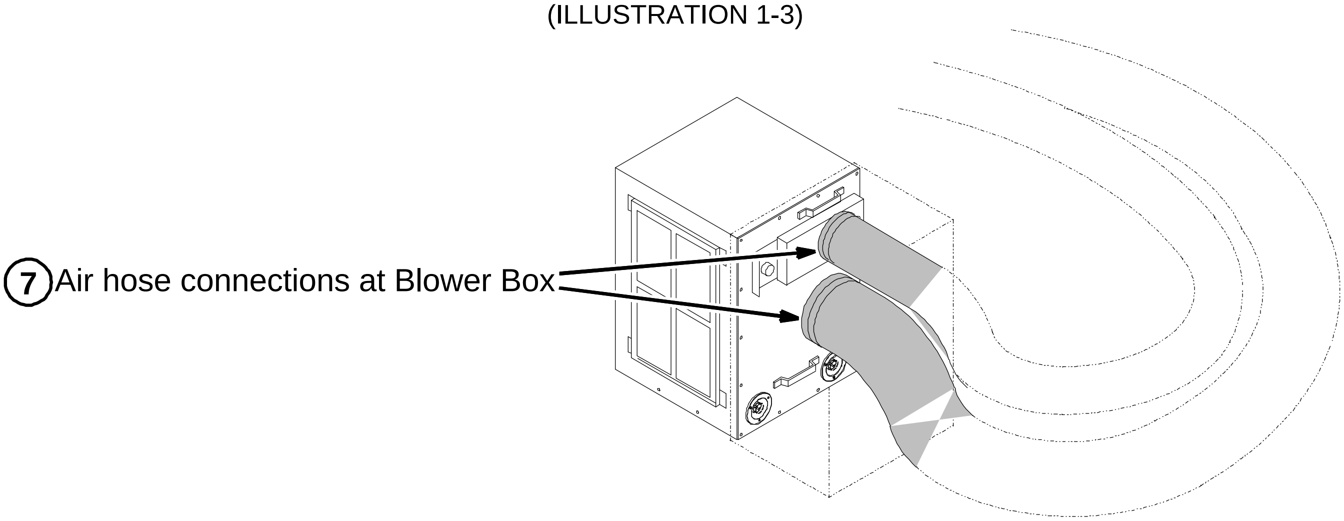
7. The air hose connection at the blower box. (See Illustration 1-4)



REAR OF MAGNET
(ILLUSTRATION 1-2)



REAR END BELL
(ILLUSTRATION 1-3)



BLOWER BOX
(ILLUSTRATION 1-4)

2- MRCONFIG FILE CHANGES

The location of the Bore Wall Temperature Sensor has been moved on the Signa MR/*i* & CV/*i* systems with the Wide Open Enclosure and LCC magnet. The relocation of the Bore Wall Temperature Sensor requires the Bore Wall Temperature limits to be changed in the MRconfig.cfg file.

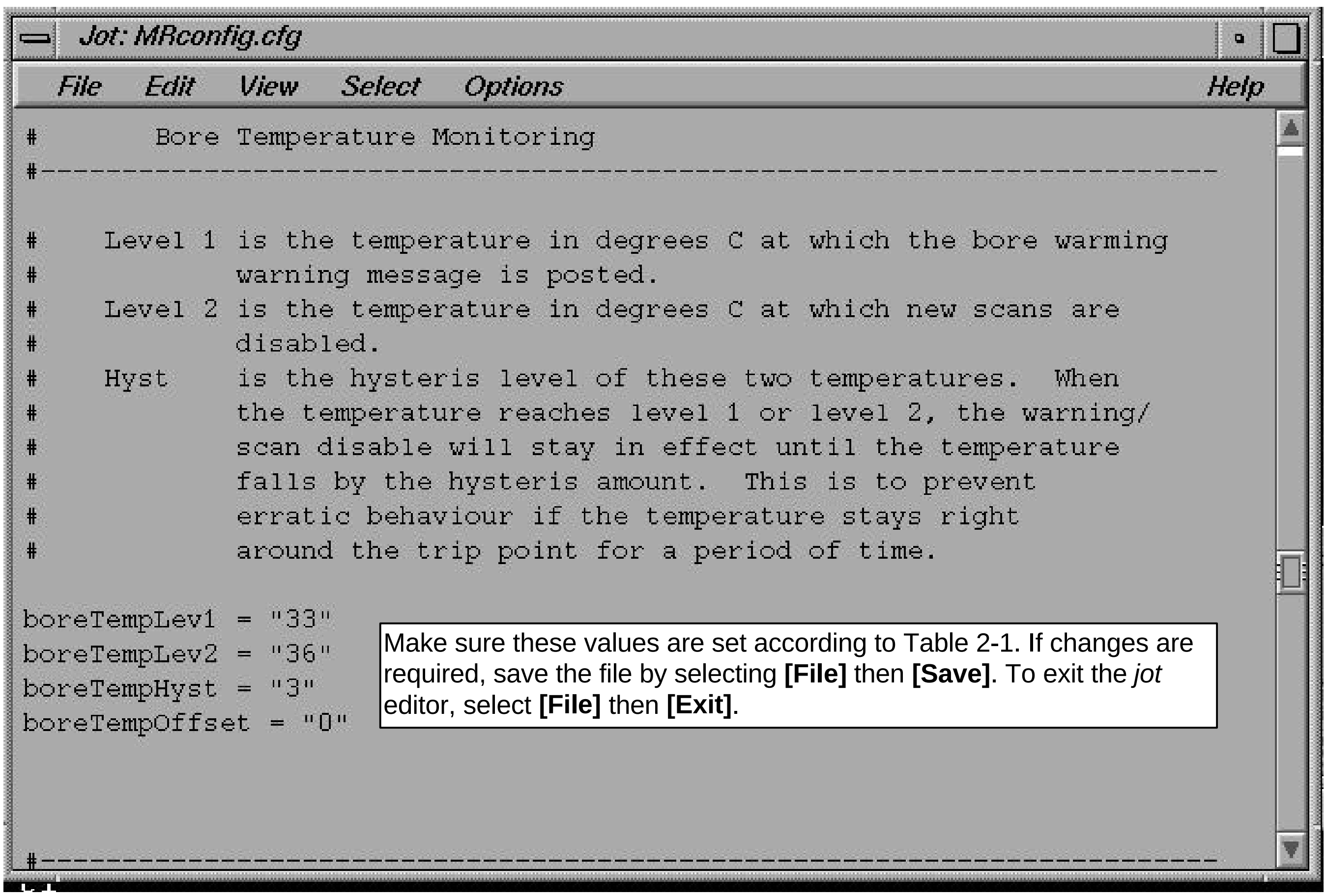
2-1 Verify Bore Temperature Trip Points

The new trip points for the Signa MR/*i* & CV/*i* Wide Open Enclosure with LCC magnet should be set according to Table 2-1. Open up a **[Cshell]** from the Service Desktop and use the *jot* file editor to view/change the MRconfig.cfg file to reflect the New Values in Table 2-1. See Illustration 2-1 for starting the *jot* editor.

TABLE 2-1
TEMPERATURE TRIP POINTS

PARAMETER	OLD VALUE	NEW VALUE
boreTempLev1	29	31
boreTempLev2	33	36
boreTempHys	2	3

STARTING JOT EDITOR
ILLUSTRATION 2-1



JOT EDITOR
ILLUSTRATION 2-2

3- TROUBLESHOOTING

If the system has **LARGE** air leaks that cannot be resolved by simple hardware or seal adjustment, contact the OnLine Center for further direction.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Nov. 25, 1998	R. Kaufman	Preliminary Release for Maestro II.